

Project Overview: Avionic Runway ATC Design

Hello! In this portfolio I have designed an avionics based system employing both Inter task communication along with priority inheritance as a demonstration of parallel compatibility of these two systems which I don't believe was done too in depth. This application follows the ease of application 1 implemented with the with the full capacity of the courses content to provide us with a full experience of how these coding architectures can truly make a difference.

In this system I have created a queue and emergency systems for pilots landing at small airports that have little to no air traffic control capabilities (ATC). As we know from delays across the US in flight terms, ATC is hard to come by so I designed this system with that reasonable assumption in mind. What if we needed to operate in a world without them? This is the problem this project looks into.

First we see the ITC model showing a synchronization in runway LEDs, in green, and a pilot cockpit LED showing the queue status of the availability of the runway. This is interacted with a shared message system pinned to core 0, with the message being shared being the input of a Pilot operated button that can request queuing. In this was of First in first out FIFO architecture, we can achieve actionable implementation of this system. For sake of robustness, I've also added a serial monitor that is present to the pilots in the cockpit via an in-room LCD display. This overlays the priority inheritance portion of this demonstration handled by a mutex that raises the priority of the lower task to be able to complete its maintenance work. When inheritance is turned on, this is what we expect. 1.) Any input of the distress signal is instantly shown in the pilot queue and runway led confirmation, the blocking only occurs in the serial monitors data that gets overlaid onto the [LCD](#). When priority inheritance is turned off, we get an expected result of unbounded task limit times that makes operating the system dangerous with this induced failure. While the mere printing of this emergency task is a soft deadline, if this induced failure were to occur for a much more hard hitting task like lifting the wheels upon landing, then a serious task failure would occur, and this is seen in the Liu Layland Reaching its upperbound of task schedulability when $PI=0$. Many instances of our medium priority tasks (Telemetry data) can run making stall emergency serial printing unable to be predicted.